

TypeScript Full Course Outline

Module 1: Introduction to TypeScript

1. What is TypeScript?
2. Differences Between JavaScript and TypeScript
3. Installing TypeScript & Setting Up the Environment
4. First TypeScript Program
5. Compiling TypeScript Code
6. Understanding TypeScript Configuration (tsconfig.json)

Module 2: Type System & Basic Types

1. Primitive Types: string, number, boolean
2. Special Types: null, undefined, any, unknown, void, never
3. Type Assertions
4. Type Inference
5. Working with Enums

Module 3: Working with Functions

1. Function Types & Signatures
2. Optional & Default Parameters
3. Function Overloading
4. Arrow Functions & this Keyword

Module 4: Objects, Interfaces, and Type Aliases

1. Defining Objects & Interfaces
2. Readonly & Optional Properties
3. Extending Interfaces
4. Type Aliases vs Interfaces
5. Intersection & Union Types

Module 5: Classes & Object-Oriented Programming in TypeScript

1. Class Basics: Properties & Methods
2. Access Modifiers (public, private, protected)
3. Getters & Setters
4. Static Properties & Methods

5. Abstract Classes & Interfaces
6. Implementing Inheritance
7. Polymorphism in TypeScript

Module 6: Generics & Advanced TypeScript Features

1. Understanding Generics
2. Generic Functions & Classes
3. Generic Constraints
4. Utility Types (Partial, Required, Readonly, etc.)
5. Type Guards & Type Narrowing
6. Conditional Types & Mapped Types

Module 7: Modules & Namespaces

1. Understanding Modules
2. Importing & Exporting Modules
3. Default vs Named Exports
4. Namespaces in TypeScript

Module 8: Working with Third-Party Libraries

1. Using DefinitelyTyped (@types)
2. Installing and Using Type Definitions
3. Working with External APIs
4. Creating Custom Type Definitions

Module 9: Error Handling & Debugging

1. Handling Errors with Try-Catch
2. Custom Error Classes
3. Using Debugging Tools in TypeScript

Module 10: TypeScript with Frontend & Backend Frameworks

1. Using TypeScript with React
2. TypeScript with Node.js & Express
3. Working with TypeORM & Prisma for Databases
4. TypeScript in Next.js
5. Integrating TypeScript with Redux

Module 11: Testing in TypeScript

1. Introduction to Unit Testing in TypeScript
2. Using Jest for Testing
3. Writing Type-Safe Tests
4. Mocking & Spying in TypeScript

Module 12: TypeScript Best Practices & Performance Optimization

1. Best Coding Practices for TypeScript
2. Improving TypeScript Performance
3. Working with Large-Scale TypeScript Projects
4. TypeScript Code Optimization
5. Setting Up Linting & Formatting (ESLint & Prettier)

Final Project: Full-Stack TypeScript Application

- Project Overview: Build a full-stack project using TypeScript, React (Frontend), Node.js + Express (Backend), and PostgreSQL/MongoDB (Database).
- Features:
 - User Authentication (JWT & Session-based Auth)
 - CRUD Operations with REST API
 - State Management using Redux Toolkit
 - Role-Based Access Control
 - Type-safe API calls using axios & TypeScript interfaces
 - Frontend Validation with React Hook Form & TypeScript
 - Deployment using Vercel (Frontend) & Railway/Heroku (Backend)

Conclusion & Next Steps

- Revisiting Key Concepts
- Resources for Further Learning
- Real-world Applications of TypeScript
- Open Source Contributions & TypeScript Community